Erfan Najafian

Master's Student in Theoretical Physics

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LinkedIn | GitHub

Research Profile

Master's student in Theoretical Physics at Sharif University of Technology with a focus on High-Energy Physics. My primary research interests lie at the intersection of Quantum Gravity, General Relativity, Mathematical Physics, and Cosmology. I am particularly interested in gravitational theories, AdS/CFT, quantum field theory, and mathematical structures underlying fundamental physics. My academic background includes advanced coursework and independent study in General Relativity, Quantum Field Theory, Differential Geometry, Topology, and related areas of theoretical physics.

Education

Sharif University of Technology — **Oct 2025 – Jul 2027**
M.Sc. in Physics — High Energy Physics — *Tehran, Iran*

- **Current GPA:** 3.23/4.00
- **Research:** *Application of the Method of Regions in AdS/CFT* — Ongoing
- **Supervisor:** Dr. Arash Ardehali

Ferdowsi University of Mashhad — **Oct 2021 – Feb 2024**
B.Sc. in Physics — *Mashhad, Iran*

- **GPA:** 3.65/4.00
- **Academic Standing:** Ranked 1st among incoming Physics students
- **Graduation:** Completed in seven semesters
- **B.Sc. Project:** *Thermodynamics of Black Holes*
- **Supervisor:** Dr. Ahmad Ghodsi

Research & Academic Projects

Application of the Method of Regions in AdS/CFT — *2026 – Present*
M.Sc. Research Project — Sharif University of Technology — *Supervisor: Dr. Arash Ardehali*

- Ongoing research project in High-Energy Theoretical Physics focusing on applications of the method of regions in the context of AdS/CFT.

Three-Dimensional Topologically Massive Gravity — *2025*
Academic Project — Advanced Topics in General Relativity — *Supervisor: Dr. Mashhoun*

- Independent written academic project on three-dimensional topologically massive gravity, completed as part of an advanced General Relativity course.

Thermodynamics of Black Holes — *2023–2024*
B.Sc. Academic Project — Ferdowsi University of Mashhad — *Supervisor: Dr. Ahmad Ghodsi*

- Undergraduate project focused on the thermodynamic properties of black holes and their theoretical description.

Honors & Awards

- **Ranked 4th** in the National Ph.D. Entrance Examination in Physics, 2025.
- **Ranked 13th** in the National Student Physics Olympiad, 2024.
- **Ranked 5th** in the National M.Sc. Entrance Examination in Physics, 2024.
- **Ranked 1st among incoming Physics students** at Ferdowsi University of Mashhad.

Teaching Experience

Teaching Assistant / Problem-Solving Instructor — **2021 – 2024**
Ferdowsi University of Mashhad — *Mashhad, Iran*

- **Physics I** — Problem-solving sessions, Dr. Roshan
- **Physics II** — Problem-solving sessions, grading, and partial instruction, Dr. Ghodsi
- **Mathematics II** — Problem-solving sessions, Dr. Parvizi
- **Differential Equations** — Problem-solving sessions, Dr. Mortazavi
- **Electromagnetism I & II** — Problem-solving sessions, Dr. Javidan
- **Mathematical Physics I** — Problem-solving sessions, Dr. Khazaei
- **Mathematical Physics III** — Problem-solving sessions, grading, and examination design, Dr. Ghodsi
- **Physics I** — Problem-solving sessions, Dr. Taghavi

Official Physics Olympiad Team Instructor **2025**
Ferdowsi University of Mashhad *Mashhad, Iran*

- Official instructor for the Ferdowsi University student Physics Olympiad team, with formal institutional documentation. Four of five team members were subsequently admitted through the relevant national selection.

Private Instructor in Gravitation **2023**
Private Academic Instruction *Mashhad, Iran*

Academic Presentations & Workshops

- **General Relativity Presentations** — Three academic presentations at Ferdowsi University of Mashhad, 2024–2025.
- **Undergraduate Physics Workshops** — Two workshops for undergraduate students covering the development of early quantum mechanics, including discussions of Schrödinger's 1926 work and Heisenberg's 1925 work.

Selected Coursework

Quantum Field Theory	General Relativity	Advanced Quantum Mechanics
Statistical Field Theory	Particle Physics	Cosmology
String Theory	Electrodynamics	Mathematical Physics

Research Interests

Quantum Gravity
Mathematical Physics
Quantum Field Theory
AdS/CFT
General Relativity
Cosmology
Gravitational Theories

Technical Skills

Programming Python (Intermediate), MATLAB (Intermediate)
Typesetting LaTeX (Advanced)
Other Academic writing, mathematical typesetting, theoretical calculations

Languages

Persian — Native
English — B2

IELTS examination planned; target overall band: 7.0.

Academic References

Dr. Mahmood Roshan
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Dr. Ahmad Ghodsi
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Dr. Hamidreza Afshar
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